

SUBJECT: MATH CLASS: VIII
CHAPTER : PROPORTIONAL REASONING-2

DAY - 1

3.1 Proportionality - A Quick Recap To begin, let's revisit the basics of ratios. When two or more related quantities change by the same factor, we call that relationship a proportional relationship.

The Idli Batter Challenge

Viswanath and Puneet are making idli batter using different quantities:

- **Viswanath:** 6 cups of rice and 3 cups of urad dal (Ratio 6:3).
- **Puneet:** 4 cups of rice and 2 cups of urad dal (Ratio 4:2).

Figure it Out Would their idlis taste the same? We use the cross-multiplication method to verify:

- $6 \times 2 = 12$
- $3 \times 4 = 12$

Since the products are the same, the ratios are proportional.

CW: Practice cross-multiplication with ratios 5:10 and 2:4.

HW: Recall the general rule $a \times d = b \times c$.

DAY - 2

3.2 Ratios in Maps Have you noticed a ratio like 1:60,00,000 in the corner of a map? This is called a Representative Fraction (RF).

- **Definition:** RF shows the ratio between a distance on the map and the corresponding actual distance on the ground.
- **Example:** 1 cm on the map is equivalent to a geographical distance of 60,00,000 cm.
- **Conversion:** 60,00,000 cm is 60 km.

Activity: Use a ruler to find the distance between Bengaluru and Chennai on the map (page 56), then use the RF to find the actual geographical distance.

CW: Page 56 - Map distance calculations. **HW:** Verify if different maps of the same area give the same geographical distance.

DAY - 3

3.3 Ratios with More than 2 Terms Ratios can have many terms if each quantity changes by the same factor to maintain proportionality.

The Spice Mix Model Viswanath's spice mix ratio (Coriander: Chillies: Toor Dal: Fenugreek) is 8:4:2:1.

[Insert Picture: Four small bowls containing spice seeds labeled with their respective parts: 8, 4, 2, and 1]

Problem Solving: Puneet only has 2 red chillies (half of Viswanath's 4). To keep the taste the same, he must halve everything else:

- Puneet should add 4 spoons of coriander, 2 red chillies, 1 spoon toor dal, and 0.5 spoon fenugreek.

- **New Ratio:** 4:2:1:0.5.
- **Proportionality:** 8:4:2:1 :: 4:2:1:0.5.

CW: Practice reducing multi-term ratios.

HW: If $a:b:c:d :: p:q:r:s$, then $\frac{a}{p} = \frac{b}{q} = \frac{c}{r} = \frac{d}{s}$.

DAY - 4

3.4 Dividing a Whole in a Given Ratio To divide a total amount among multiple parts, we add the terms and divide the whole by the sum.

Example 1: Purple Paint Ratio: Red: Blue: White :: 2:3:5. If Yasmin has 10 litres of white paint (the 5 parts):

- 1 part = $\frac{10}{5} = 2$ litres.
- Red = 2 parts $\times 2 = 4$ litres.
- Blue = 3 parts $\times 2 = 6$ litres.
- **Total Volume:** $4 + 6 + 10 = 20$ litres.

CW: Page 58 - Example 2 (Concrete mixture bags). **HW:** Figure it Out (Page 60) - Question 1 (Cricket coach schedule).

DAY - 5

Dividing Units in Ratio (Practical Application)

Continuing our work on dividing wholes, let's look at larger construction needs.

Example 3: Concrete Mixture For 110 units of concrete in the ratio 1:1.5:3:

- Sum of parts = $1 + 1.5 + 3 = 5.5$.
- Number of times needed = $\frac{110}{5.5} = 20$.
- Cement: $1 \times 20 = 20$ units.
- Sand: $1.5 \times 20 = 30$ units.
- Gravel: $3 \times 20 = 60$ units.

CW: Page 59 - Example 4 (Purple paint ml). **HW:** Figure it Out (Page 60) - Question 2 (Library books).

DAY - 6

3.5 A Slice of the Pie Pie charts show different proportions of a whole, helping us quickly interpret data.

How to Construct a Pie Chart:

1. **Calculate Total Angle:** The total angle in a circle is 360° .
2. **Angle Calculation:** To mark a slice, the angle must be proportional to the value.
 - **Formula:** $Angle = \frac{Value}{Total} \times 360^\circ$.
3. **Simplifying Ratios:** Divide all terms by their HCF to get the simplest form for easier calculation.

CW: Calculate angles for Grade Distribution (40 students) on page 61. **HW:** Figure it Out (Page 62) - Question 1 (Favourite season).

DAY - 7

3.6 Inverse Proportions When two ratios are proportional such that one value increases while the other decreases, it is an inverse proportion.

The Speed-Time Challenge Consider traveling from Lucknow to Kanpur.

- **Motorcycle:** 30 km/h takes 3 hours.
- **Car:** 60 km/h takes 1.5 hours. When the speed increases, the time taken decreases.

[Insert Picture: A table showing Speed vs. Time with arrows showing speed multiplying while time divides]

Inverse Proportion Rule: Two quantities x and y vary in inverse proportion if $xy = k$ (a constant).

- Speed $\times$ Time = Distance (Constant).
- $x_1/x_2 = y_2/y_1$.

CW: Page 64 - Mode of transport table analysis. **HW:** Page 65 - Figure it Out, Question 1 (Identify inverse proportions).

DAY - 8

Inverse Proportions in Labor and Resources

Inverse proportion often applies to people working or sharing supplies.

Example 3: Road Work

- 20 workers take 4 days.
- How many days will 10 workers take?
- $20 \times 4 = 10 \times y_2$.
- $y_2 = 8$ days. When the number of workers is halved, the days double.

Example 5: Food Provisions

- 80 students have food for 15 days.
- If 20 more join (100 total), how long will it last?
- $80 \times 15 = 100 \times x \rightarrow x = 12$ days.

CW: Page 66 - Example 4 (Tank pumps). **HW:** Page 67 - Figure it Out, Question 3 (Water tank supply).

DAY - 9

Working Together (Complex Proportions)

What happens when people with different speeds work together?

Example 6: Cutting Vegetables

- Ram takes 1 hour (1 unit/hour).
- Shyam takes 1.5 hours ($\frac{2}{3}$ units/hour).
- Together in 1 hour: $1 + \frac{2}{3} = \frac{5}{3}$ units of work.
- To complete 1 unit together, the time x is found by $\frac{5}{3} : 1 :: 1 : x$.
- $x = \frac{3}{5}$ hours.

Activity: Practice the "Together" rule for filling tanks with different sized pumps.

CW: Page 68 - Question 6 (Painting a fence). **HW:** Page 68 - Question 10 (Pump filling tank).

DAY - 10

Summary and Practical Review

Recapping the key differences between direct and inverse proportionality.

Key Summary Points:

- **Direct Proportion:** Both change by the same factor; $\frac{x}{y} = k$.
- **Inverse Proportion:** One changes by n , the other by $\frac{1}{n}$; $x \times y = n$.
- **Multi-term Ratios:** $x \times \frac{p}{(p+q+r+\dots)}$ finds a specific part of a whole.

[Insert Picture: A two-column chart labeled "Direct" vs "Inverse" with real-world icons (e.g., price vs quantity vs speed vs time)]

Final Challenge Task:

- Study the "Modes of Transport" pie chart on page 68.
- Determine the most common mode and total survey participants.

CW: Page 68 - Questions 11 and 12 (Machines and Car speed). **HW:** Review all Summary points on page 69.

PAGE – 60

CW – Q1 , Q3 , Q5

HW – Q2, Q4

PAGE – 65

CW – Q1 – II , Q2

HW – Q1 – I , III

PAGE – 62

CW – Q1 , Q3

HW – Q2

PAGE – 67

CW – Q2, Q3 , Q4 , Q6, Q8 , Q9 , Q11 , Q12

HW – Q1, Q5 , Q7 , Q10